

NEWVISION

Functional Specification

Roles, workflows, lifecycles, and QA-oriented behavior

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1. Purpose and audience

This document describes what the NewVision product does: features, role behavior, status lifecycles, and cross-module workflows. Technical deploy and debug details are in NewVision_Developer_Handoff.docx and docs/DEVELOPER_HANDBOOK.md.

Source of truth Server validators, service throw paths, Prisma enums, and e2e specs override help articles and this doc when they disagree.

2. Functional inventory

Module	Primary roles	Route prefix
Dashboard	All	/
Assets	IT Admin+, Employee (scoped)	/assets
Employees	IT Admin+, Manager, Employee	/employees
Asset requests	Employee, Manager, IT Admin	/requests
Maintenance	IT Admin, IT Support (module)	/maintenance
Support tickets	All (scoped), IT Support manage	/tickets
Procurement	IT Admin, Manager request (module)	/procurement/*
Chat	IT staff (module)	/chat
Settings / workspace	Super Admin, IT Admin	/settings
Public QR scan	Anyone with code	/scan/:code

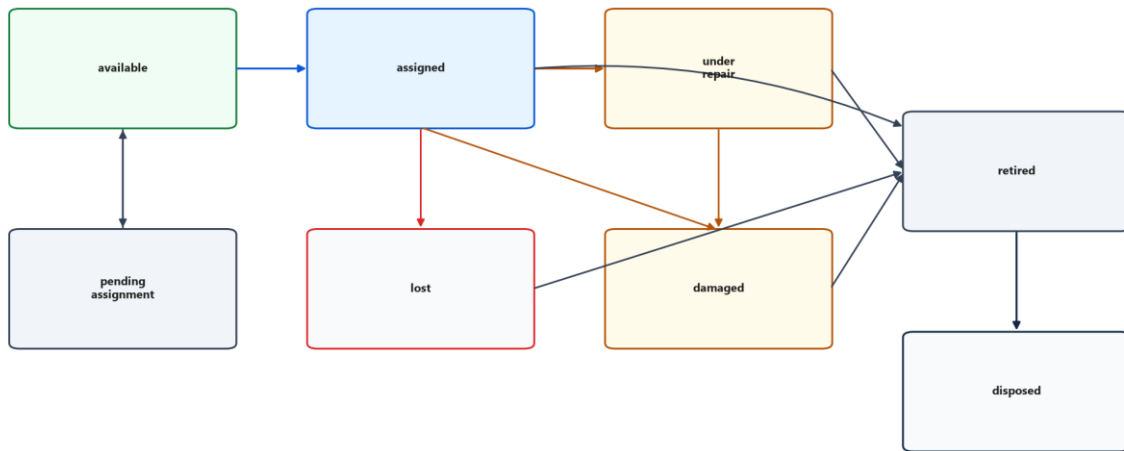
3. Roles and permissions

Full matrix: docs/ROLE_PERMISSION_MATRIX.md. Summary:

Role	Summary
SUPER_ADMIN	All permissions including user:manage and asset:delete
IT_ADMIN	Full IT operations except user:manage and asset:delete
IT_SUPPORT	Tickets, maintenance, read assets/employees, reports
MANAGER	Team scope, approve requests, procurement request, reports
EMPLOYEE	My kit, raise requests, report issues, own tickets

Plan gating: Starter vs Team trial — procurement, chat, maintenance hidden when modules false (tenancy/plans.ts, access.ts filterNav).

4. Status lifecycles



available also → under_repair (spare in storage). lost → available or retired. damaged → under_repair, available, or retired. Transfer is an action, not a status.

Same-status is idempotent. disposed is terminal. Hard-delete is Super Admin only on retired/disposed.

Figure. Asset status transitions (lifecycle.ts).

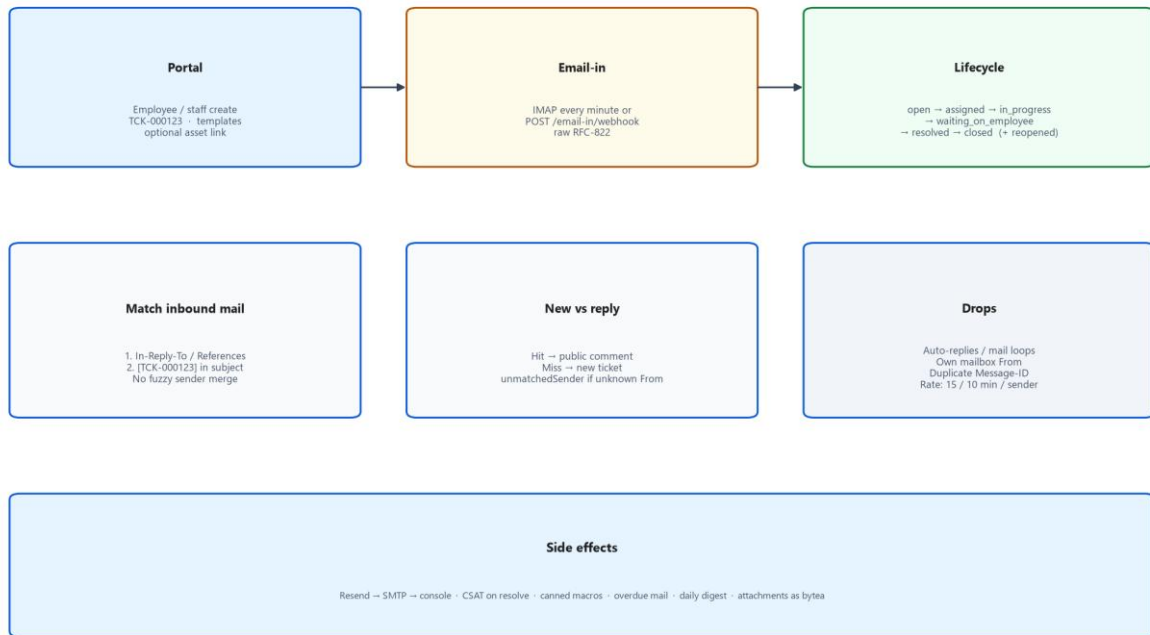


Figure. Ticket channels and lifecycle (simplified).

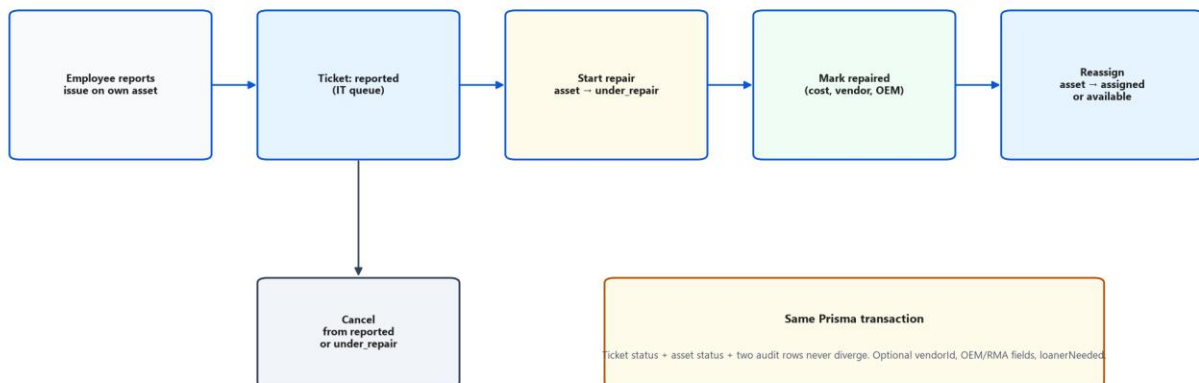
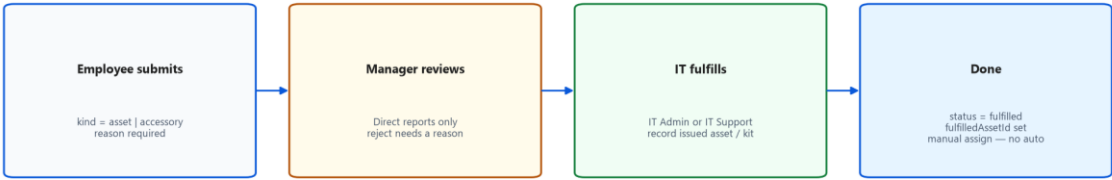


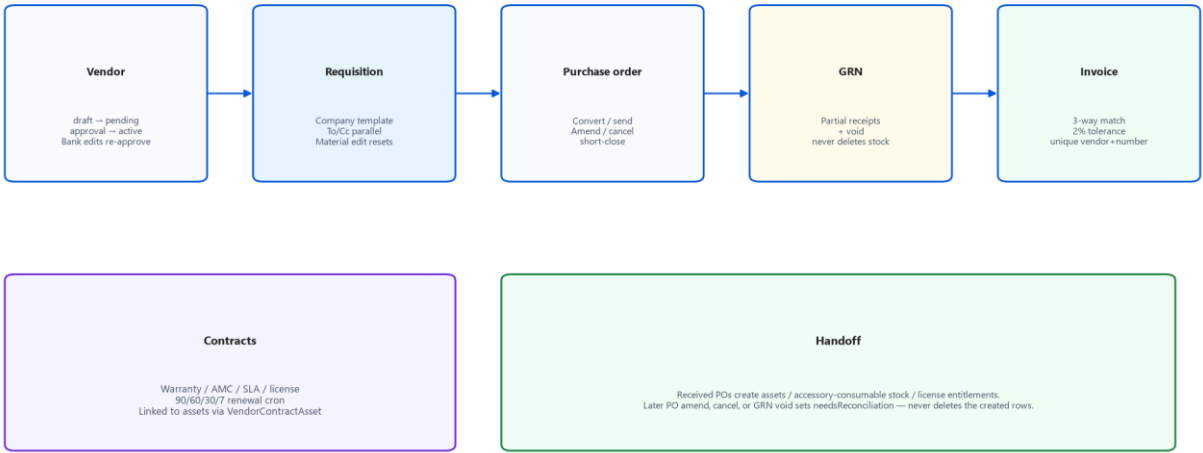
Figure. Maintenance ticket coupled to asset status.



States: pending → approved | rejected → fulfilled. This is not a workflow engine. Managers never fulfill. Employees cannot see other people's requests.

IT Support was given fulfill permission so the Requests nav item is not a dead end (Prompt 32).

Figure. Asset request: pending → approved/rejected → fulfilled.



RBAC: Super Admin / IT Admin own Vendors, POs, Contracts. Managers raise and see team requisitions only. IT Support gets /vendors/options (names, no bank) for repair pickers.

Team plan module. Trial tenants must finish or skip the first-hour checklist before procurement APIs open.

Figure. Procurement chain.



Synchronous POST `/api/import/assets/employees` still exists for small files. Jobs live under Settings, not a new sidebar item.

Reconciliation is a separate manual CSV set-diff (employeeCode/email or assetCode/serialNumber) — not live AD/HR sync.

Figure. Import job states.

5. Cross-module workflows

Detailed steps: docs/WORKFLOWS.md. Highlights:

1. New hire: employee record → user login → assign asset or issue kit → employee sees My kit.
2. Broken device: employee ticket and/or IT maintenance; asset may enter under_repair.
3. Request: employee submits → manager approves/rejects (reason required) → IT fulfills.
4. Procurement: requisition approvals → PO → goods receipt → assets/invoices.
5. Email-in: reply with [TCK-...] adds comment; unmatched sender flagged.
6. QR audit: public scan card; staff audit stamps drive dashboard KPIs.

6. Notifications and automation

Trigger	Channel	Source
Asset request to manager	In-app	asset-requests.service.ts
Request approved/rejected	In-app	asset-requests.service.ts
Warranty 90/60/30 days	In-app + email	warranty-alert.service.ts
Ticket digest / overdue	Email	tickets.digest.ts
Contract renewal	Cron	contracts.service.ts
IMAP poll	Ticket comment	email-inbox.service.ts

7. Known functional gaps

Severity	Issue
HIGH	/signup and /trust pages exist but are not registered in App.tsx (REFERENCED only).
MEDIUM	Help articles say no public signup; login page promotes trial signup.
MEDIUM	Help copy may lag code — verify against services and e2e.

8. Glossary (selected)

Term	Meaning
My kit	Employee nav label for assigned devices
Team devices	Manager view of assets
TCK-#####	Ticket token in email subject
Starter / Team	Tenant plan tiers
unmatchedSender	Email-in ticket from unknown address

9. Related documents

- docs/FUNCTIONAL_SPECIFICATION.md — full markdown spec
- docs/WORKFLOWS.md — workflow diagrams (Mermaid in repo)
- docs/ROLE_PERMISSION_MATRIX.md — permission tables
- NewVision_System_Flow_and_Usage_Guide.docx — end-user flows
- NewVision_Developer_Handoff.docx — technical architecture